



Analysis of water wave scattering by a submerged perforated reef ball using multipole method

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Received: 7 January 2019 / Accepted: 6 September 2019
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Abstract Based on linear potential theory, this study develops an analytical solution to water wave scattering by a submerged perforated reef ball. The series solutions for velocity potentials in the external and internal fluid domains of the perforated reef ball are developed using the multipole expansions and the separation of variables, respectively. The unknowns in the series solutions are determined by matching the porous boundary condition on the perforated reef ball. Also a numerical solution using three-dimensional multi-domain boundary element method (BEM) is developed. The special case of an impermeable reef ball is also examined. The calculation methods for the horizontal and vertical wave forces acting on the reef ball and the free surface elevation around the reef ball are given. The convergence of the calculation results is very rapid, and the results with five-figure accuracy are obtained using a few truncated terms in the series solution. The results by the analytical solution agree well with that by the multi-domain BEM solution. Typical calculation examples are presented to examine the effects of reef ball radius, porous effect parameter of reef ball and incident wave frequency on the wave forces and the free surface elevation. The results of analysis are valuable for engineering design

and application of reef balls. The present analytical solution gives a new reliable benchmark for three-dimensional numerical models to wave interactions with perforated structures. Moreover, better understanding on wave scattering by perforated structures can be obtained during the analytically solving procedure.

Keywords Wave scattering · Perforated reef ball · Multipole expansion · Wave forces · Free surface elevation

1 Introduction

Artificial structures have been widely used in efforts to increase marine fish abundance and diversity, restore damaged coral reefs, prevent coastal erosion and provide human activities protection. The shape of artificial structures varies with different functionality. The hemispherical shape structures, which are so-called reef ball, have been used as artificial reefs to provide marine life peaceful shelter [1, 2]. This type of structure can be regarded as a perforated hemispherical thin shell. Compared with the rectangular reef, polyhedral reef and cylindrical reef, the perforated reef ball has the merit of almost zero overturning moment because the dynamic pressure passes through the spherical center. Besides, the wave forces acting

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on this type of reef are reduced effectively due to the porosity of the thin shell. On the other hand, a mass of perforated reef balls with different appropriate layouts can be used as submerged breakwaters to provide opportunities for environmental enhancement, aesthetics and wave energy dissipation [3–6]. Since the reef balls are utilized in practice from different fields, they have received considerable attentions from researchers and engineers in terms of the safety, stability, cost-efficiency, environmental friendliness and other aspects. It is very necessary to assess their hydrodynamic performance. This paper will develop an analytical solution to water wave scattering by a single submerged perforated reef ball, and give an insight regarding the hydrodynamic performance of this structure. This should be of significance for engineering design and application of reef balls and similar structures. Before conducting present investigation, a brief review on wave interaction with spherical shape structures is presented.

Plenty of studies for water wave interaction with the spherical structures have been carried out for a few decades. Many studies are based on the potential flow theory, and mainly focused on examinations of wave forces acting on the structures, added mass and radiation damping, free surface elevation and velocity flow field near the structures, all of which have significant influence on safety and stability of structures and economic cost. Srokosz [7] regarded a submerged tethered sphere as a wave power absorber, and analysed the effects of the added mass and radiation damping on the power absorption performance. Wu and Eatock Taylor [8] developed analytical solutions to estimate the dynamic forces acting on submerged spheres moving along a circular path. Miloh [9] examined the impulsive oblique motion of a spherical structure in close proximity and below the free surface in infinite depth water. Rahman [10] solved the wave diffraction problem for a submerged floating sphere in finite depth water, and paid his attention to the pressure on body surface, the free surface elevation and the fluid velocity field. Ursell [11] developed analytical solutions to the wave radiation caused by a submerged sphere moving along a canal in infinite depth water. More studies on the wave interactions with spherical structures can be found in Wang [12], Linton [13], Wu [14], Cadby and Linton [15] and Liu et al. [16]. In the aforementioned studies, a single submerged sphere has been examined.

In addition, Wu [17] solved the water wave diffraction and radiation problems for a group of submerged spheres.

The structures in above studies are complete spheres. Sometimes, the spherical body may be not a complete sphere, such as a hemisphere and a sphere with an opening hole. Based on the study of Havelock [18], Hulme [19] further investigated the added mass and radiation damping coefficients due to the heave and surge motions of a floating hemisphere, and gave a derivation for the long- and short-wave approximate values of these coefficients. Dong et al. [20] studied the interaction between surface waves and a submerged floating sphere with an opening hole, and examined the wave forces on the sphere. Chapman [21] developed an analytical solution for water wave scattering by a hemispherical seamount, and calculated the “scattering coefficient”. However, in the study of Chapman [21], the submerged hemispherical structure is impermeable and both the wave forces acting on it and the free surface elevation were not examined. To the best of authors’ knowledge, the hydrodynamic performance of a submerged perforated hemispherical structure (a perforated reef ball) has not been analytically examined so far. The main objective of this study is to develop an analytical solution to better understand the hydrodynamic performance of the submerged perforated reef ball, which is idealized as a uniformly perforated hemispherical shell. The multipole expansion method [22–24] will be utilized in the solution procedure. Also, a numerical solution using three-dimensional multi-domain boundary element method (BEM [25–31]) will be developed for cross-checking of the analytical solution.

The governing equation and boundary conditions for the present three-dimensional wave scattering problem are given in the following Section. In Sect. 3, the multipoles with singularity on the water bottom are briefly deduced, and then the solution procedure using multipole expansions is detailed. The calculation methods of wave forces acting on the perforated reef ball and the free surface elevation near the reef ball are presented. In Sect. 4, the special case of an impermeable reef ball is considered. In Sect. 5, the convergence of the analytical solution is examined. The calculation results by the analytical solution are compared with that by an independently developed three-dimensional multi-domain BEM solution, which is introduced in “Appendix”. The basic features of the

wave forces and the free surface elevation are investigated. In Sect. 6, the limitation of the present solution is stated. Finally, main conclusions of this study are drawn.

2 Mathematical formulations

The water wave scattering by a submerged perforated reef ball with radius a in finite depth water is considered here. Figure 1 shows the idealized sketch: (a) is the side view and (b) is the top view. The depth of water is h . A three-dimensional Cartesian coordinate system, with the origin fixed on the undisturbed free surface, the z -axis taking vertically upwards and passing through the center of the reef ball and the x -axis pointing the right hand side, is used for the mathematical descriptions. A spherical coordinate system is defined with

$$\begin{aligned} x &= r \sin \theta \cos \beta, & y &= r \sin \theta \sin \beta, \\ z + h &= r \cos \theta. \end{aligned} \quad (1)$$

It is assumed that the fluid is inviscid and incompressible and its motion is irrotational. Assuming the incident wave with angular frequency ω propagates along the positive x -direction, a fluid velocity potential $\Phi(x, y, z, t)$ can be used to describe the whole fluid motion. After the separation of the time factor $e^{-i\omega t}$, the velocity potential can be written as

$$\Phi(x, y, z, t) = \text{Re} \left[-\frac{igH}{2\omega} \phi(x, y, z) e^{-i\omega t} \right], \quad (2)$$

where Re denotes the real part of function; $i = \sqrt{-1}$; g is the gravitational acceleration; H is the incident

wave height; $\phi(x, y, z)$ is the reduced spatial velocity potential and t is the time.

The spatial velocity potential ϕ satisfies Laplace equation

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0. \quad (3)$$

Besides, the velocity potential satisfies the free surface condition, the impermeable water bottom condition and the Sommerfeld radiation condition:

$$\frac{\partial \phi}{\partial z} = K\phi, \quad K = \frac{\omega^2}{g}, \quad z = 0, \quad (4)$$

$$\frac{\partial \phi}{\partial z} = 0, \quad z = -h, \quad (5)$$

$$\lim_{\tilde{r} \rightarrow \infty} \sqrt{\tilde{r}} \left(\frac{\partial}{\partial \tilde{r}} - ik \right) (\phi - \phi_0) = 0, \quad (6)$$

where $\tilde{r} = (x^2 + y^2)^{1/2}$; k is the incident wave number and the positive real root of the relationship

$$K = k \tanh(kh); \quad (7)$$

and ϕ_0 is the incident wave velocity potential and has the form of

$$\phi_0 = \frac{\cosh(k(z+h))}{\cosh(kh)} e^{ikx}. \quad (8)$$

When considering the wave scattering problem, the thickness of the hemispherical shell is assumed to zero as it is very small compared with both the radius of the reef ball and the incident wavelength. Then, the whole fluid domain is divided into two regions by the shell: region 1, the exterior region of the perforated reef ball ($r > a$); region 2, the interior region ($r < a$). On the

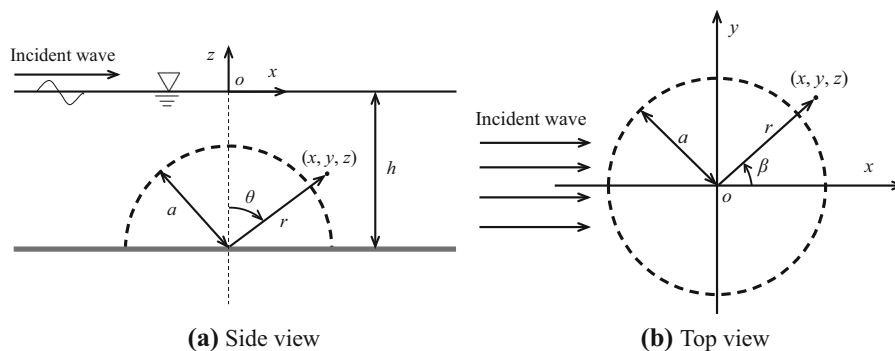


Fig. 1 Idealized sketch of water wave scattering by a submerged perforated reef ball

perforated hemispherical shell, the velocity potentials satisfy

$$\frac{\partial \phi_1}{\partial r} = \frac{\partial \phi_2}{\partial r} = ikG(\phi_2 - \phi_1), \quad r = a, \quad (9)$$

where ϕ_1 and ϕ_2 denote the velocity potentials in regions 1 and 2, respectively, and G is the porous effect parameter of the perforated hemispherical shell. Actually, Eq. (9) is the porous boundary condition on perforated plate developed by Yu [32] based on the classical porous medium model of Sollitt and Cross [33]. According to Yu [32], the porous effect parameter G is defined by

$$G = \frac{\varepsilon}{k\delta(f - is)}, \quad (10)$$

where δ is the plate thickness; and ε , f and s are, respectively, the geometrical porosity, the resistance coefficient and the inertial effect coefficient of the perforated plate (the perforated hemispherical shell). When the value of $|G|$ equals zero, the reef ball will become impermeable; however, while when the value of $|G|$ tends to infinity, the reef ball will disappear. In practical application, the value of the porous effect parameter G needs to be determined by experimental tests [34–37].

The first equal sign in Eq. (9) denotes that the normal fluid velocity is continuous on the perforated hemispherical shell. The second equal sign denotes a linear pressure drop condition, which assumes that the normal fluid velocity passing through the shell is linearly proportional to the pressure difference between both sides of the shell. It should be mentioned that differ from the linear pressure loss condition, Mei et al. [38] and Molin [39] gave different quadratic relationships between the pressure difference and the normal fluid velocity across the perforated plate, which are not considered in this study.

3 Methods of solution

In region 1, by the separation of variables, we obtain the fundamental solutions of Eq. (3) satisfying the water bottom condition in Eq. (5):

$$\frac{P_{2n+2m}^{2m}(\cos \theta)}{r^{2n+2m+1}} \cos(2m\theta), \quad m, n = 0, 1, 2, \dots, \quad (11)$$

$$\frac{P_{2n+2m-1}^{2m-1}(\cos \theta)}{r^{2n+2m}} \cos((2m-1)\theta), \quad m = 1, 2, \dots, \\ n = 0, 1, 2, \dots, \quad (12)$$

where $P_n^m(x)$ is associated Legendre function and $P_n^m(x) = (-1)^m (1-x^2)^{m/2} \frac{d^m P_n(x)}{dx^m}$, which is the same as the definitions in Linton and McIver [24] and Gradshteyn and Ryzhik [40], but differs by a factor $(-1)^m$ from Thorne [23].

According to Linton and McIver [24, Eqs. (B.63) and (B.64)], we have

$$\frac{P_{2n+2m}^{2m}(\cos \theta)}{r^{2n+2m+1}} = \frac{1}{(2n)!} \\ \times \int_0^\infty \mu^{2n+2m} e^{-\mu(z+h)} J_{2m}(\mu R) d\mu, \quad z > -h, \quad (13)$$

$$\frac{P_{2n+2m-1}^{2m-1}(\cos \theta)}{r^{2n+2m}} = -\frac{1}{(2n)!} \\ \times \int_0^\infty \mu^{2n+2m-1} e^{-\mu(z+h)} J_{2m-1}(\mu R) d\mu, \quad z > -h, \quad (14)$$

where $J_m(x)$ is the first kind Bessel function of order m , and $R = r \sin \theta$. Therefore, the multipoles can be constructed as $\phi_{n+m}^{m,+} \cos(2m\beta)$ and $\phi_{n+m}^{m,-} \cos((2m-1)\beta)$, and $\phi_{n+m}^{m,+}$ and $\phi_{n+m}^{m,-}$ are respectively written as

$$\phi_{n+m}^{m,+} = \frac{P_{2n+2m}^{2m}(\cos \theta)}{r^{2n+2m+1}} + \frac{1}{(2n)!} \\ \times \int_0^\infty A(\mu) \mu^{2n+2m} \cosh(\mu(z+h)) J_{2m}(\mu R) d\mu, \\ m, n = 0, 1, 2, \dots, \quad (15)$$

$$\phi_{n+m}^{m,-} = \frac{P_{2n+2m-1}^{2m-1}(\cos \theta)}{r^{2n+2m}} - \frac{1}{(2n)!} \\ \times \int_0^\infty B(\mu) \mu^{2n+2m-1} \cosh(\mu(z+h)) J_{2m-1}(\mu R) d\mu, \\ m = 1, 2, \dots, n = 0, 1, 2, \dots, \quad (16)$$

where $A(\mu)$ and $B(\mu)$ are the unknown functions of μ . It is noted that both Eqs. (15) and (16) satisfy the water

bottom condition in Eq. (5). Substituting Eqs. (15) and (16) into the free surface condition in Eq. (4), respectively, we obtain

$$A(\mu) = B(\mu) = \frac{(\mu + K)e^{-\mu h}}{\mu \sinh(\mu h) - K \cosh(\mu h)}. \quad (17)$$

Based on the relationship [23]

$$e^{\pm\mu(z+h)} J_m(\mu R) = (\mp 1)^m \sum_{s=m}^{\infty} \frac{(\pm\mu r)^s}{(s+m)!} P_s^m(\cos \theta), \quad (18)$$

the multipoles in Eqs. (15) and (16) can be further written as:

$$\begin{aligned} \phi_{n+m}^{m,+} &= \frac{P_{2n+2m}^{2m}(\cos \theta)}{r^{2n+2m+1}} \\ &+ \sum_{s=0}^{\infty} r^{2s+2m} A_{ns}^{m,+} P_{2s+2m}^{2m}(\cos \theta), \end{aligned} \quad (19)$$

$$\begin{aligned} \phi_0 &= \frac{\cosh(k(z+h))}{\cosh(kh)} e^{ikR \cos \beta} = \frac{\cosh(k(z+h))}{\cosh(kh)} \sum_{m=0}^{\infty} \varepsilon_m i^m J_m(kR) \cos(m\beta) \\ &= \frac{1}{\cosh(kh)} \left\{ \sum_{m=0}^{\infty} \sum_{s=0}^{\infty} \frac{\varepsilon_m (-1)^m (kr)^{2s+2m}}{(2s+4m)!} P_{2s+2m}^{2m}(\cos \theta) \cos(2m\beta) \right. \\ &\quad \left. + \sum_{m=1}^{\infty} \sum_{s=0}^{\infty} \frac{2i (-1)^m (kr)^{2s+2m-1}}{(2s+4m-2)!} P_{2s+2m-1}^{2m-1}(\cos \theta) \cos((2m-1)\beta) \right\}, \end{aligned} \quad (24)$$

$$\begin{aligned} \phi_{n+m}^{m,-} &= \frac{P_{2n+2m-1}^{2m-1}(\cos \theta)}{r^{2n+2m}} \\ &+ \sum_{s=0}^{\infty} r^{2s+2m-1} A_{ns}^{m,-} P_{2s+2m-1}^{2m-1}(\cos \theta), \end{aligned} \quad (20)$$

where

$$A_{ns}^{m,+} = \frac{1}{(2n)!(2s+4m)!} \int_0^{\infty} \frac{\mu^{2n+2s+4m} (\mu + K) e^{-\mu h}}{\mu \sinh(\mu h) - K \cosh(\mu h)} d\mu, \quad (21)$$

$$A_{ns}^{m,-} = \frac{1}{(2n)!(2s+4m-2)!} \int_0^{\infty} \frac{\mu^{2n+2s+4m-2} (\mu + K) e^{-\mu h}}{\mu \sinh(\mu h) - K \cosh(\mu h)} d\mu. \quad (22)$$

In Eqs. (21) and (22), the path of the integral passes below the point at $\mu = k$ to satisfy the far field

radiation condition in Eq. (6). The calculation methods for this type of singular integration in Eqs. (21) and (22) are detailed in Chapman [21, Chapter 4], Linton [41, Eqs. (3.7) and (4.11)] and Liu and Li [42, Appendix].

Since the multipoles have been obtained, the velocity potential in region 1 is written as

$$\begin{aligned} \phi_1 &= \phi_0 + \frac{1}{\cosh(kh)} \\ &\times \left\{ \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} C_n^{m,+} a^{2n+2m+2} \phi_{n+m}^{m,+} \cos(2m\beta) \right. \\ &\quad \left. + \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} C_n^{m,-} a^{2n+2m+1} \phi_{n+m}^{m,-} \cos((2m-1)\beta) \right\}, \end{aligned} \quad (23)$$

where $C_n^{m,+}$ and $C_n^{m,-}$ are the unknown complex coefficients, and

in which, Eq. (18) has been used; and $\varepsilon_0 = 1$ and $\varepsilon_m = 2$ for $m \geq 1$. By the separation of variables, the velocity potential in region 2, satisfying Eqs. (3) and (5), is written as

$$\begin{aligned} \phi_2 &= \frac{1}{\cosh(kh)} \\ &\times \left\{ \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} D_n^{m,+} r^{2n+2m} P_{2n+2m}^{2m}(\cos \theta) \cos(2m\beta) \right. \\ &\quad \left. + \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} D_n^{m,-} r^{2n+2m-1} P_{2n+2m-1}^{2m-1}(\cos \theta) \cos((2m-1)\beta) \right\}, \end{aligned} \quad (25)$$

where $D_n^{m,+}$ and $D_n^{m,-}$ are the unknown complex coefficients.

Substituting Eqs. (23) and (25) into the porous boundary condition on the perforated hemispherical shell in Eq. (9), multiplying both sides of the

equations by $P_{2j+2m}^{2m}(\cos \theta) \sin \theta \cos(2m\beta)$ or $P_{2j+2m-1}^{2m-1}(\cos \theta) \sin \theta \cos((2m-1)\beta)$, and integrating with respect to β and θ from 0 to 2π and 0 to $\pi/2$, respectively, and then applying the orthogonality relationships of trigonometric function and associated Legendre function

$$\int_0^{\pi/2} P_n^m(\cos \theta) P_j^m(\cos \theta) \sin \theta d\theta = \frac{\delta_{nj}(n+m)!}{(n-m)!(2n+1)}, \quad (26)$$

where both the values of $n+m$ and $j+m$ are even numbers, and $\delta_{nj} = 0$ for $n \neq j$ and $\delta_{nj} = 1$ for $n = j$, we obtain the following four sets of linear equations:

$$C_j^{m,+} - \sum_{n=0}^{\infty} C_n^{m,+} \frac{(2j+2m)a^{2j+2n+4m+1}A_{nj}^{m,+}}{2j+2m+1} + D_j^{m,+} \frac{(2j+2m)a^{2j+2m-1}}{2j+2m+1} = \frac{2j+2m}{2j+2m+1} \frac{\varepsilon_m(-1)^m(ka)^{2j+2m}}{(2j+4m)!a}, \quad m, j = 0, 1, 2, \dots, \quad (27)$$

$$C_j^{m,+} + \sum_{n=0}^{\infty} C_n^{m,+} a^{2j+2n+4m+1}A_{nj}^{m,+} + D_j^{m,+} a^{2j+2m-2} \left(\frac{2j+2m}{ikG} - a \right) = - \frac{\varepsilon_m(-1)^m(ka)^{2j+2m}}{(2j+4m)!a}, \quad m, j = 0, 1, 2, \dots, \quad (28)$$

$$C_j^{m,-} - \sum_{n=0}^{\infty} C_n^{m,-} \frac{(2j+2m-1)a^{2j+2n+4m-1}A_{nj}^{m,-}}{2j+2m} + \frac{(2j+2m-1)a^{2j+2m-2}}{2j+2m} D_j^{m,-} = \frac{2j+2m-1}{2j+2m} \frac{2i(-1)^m(ka)^{2j+2m-1}}{(2j+4m-2)!a}, \quad m = 1, 2, \dots, \quad j = 0, 1, 2, \dots, \quad (29)$$

$$C_j^{m,-} + \sum_{n=0}^{\infty} C_n^{m,-} a^{2j+2n+4m-1}A_{nj}^{m,-} + D_j^{m,-} a^{2j+2m-3} \left(\frac{2j+2m-1}{ikG} - a \right) = - \frac{2i(-1)^m(ka)^{2j+2m-1}}{(2j+4m-2)!a}, \quad m = 1, 2, \dots, \quad j = 0, 1, 2, \dots \quad (30)$$

The system of linear equations formulated by Eqs. (27)–(30) is solved by truncating n and j to N according to the desired accuracy for each value of m (the angle β mode). Then, all the unknown coefficients in the velocity potentials are determined.

The dynamic pressure $p(x, y, z)$ is obtained by the linear Bernoulli equation

$$p(x, y, z) = \frac{\rho g H}{2} \phi(x, y, z), \quad (31)$$

where ρ is the fluid density. The wave forces acting on the reef ball are obtained by integrating the dynamic pressure over the body surface. In the spherical coordinate system in Eq. (1), unit normal vector on the reef ball surface pointing away from the reef ball is $n = (\sin \theta \cos \beta, \sin \theta \sin \beta, \cos \theta)$. Therefore, the horizontal and vertical wave forces are respectively calculated by

$$\begin{aligned} F_x &= \frac{\rho g H}{2} \iint [\phi_2(a, \theta, \beta) - \phi_1(a, \theta, \beta)] (\sin \theta \cos \beta) dS \\ &= \frac{\rho g H}{2} \iint \frac{1}{ikG} \frac{\partial \phi_2(r, \theta, \beta)}{\partial r} \Big|_{r=a} (\sin \theta \cos \beta) dS \\ &= \frac{-\pi \rho g H a^2}{3ikG \cosh(kh)} D_0^{1,-}, \end{aligned} \quad (32)$$

$$\begin{aligned}
 F_z &= \frac{\rho g H}{2} \iint [\phi_2(r, \theta, \beta) - \phi_1(r, \theta, \beta)] \cos \theta dS \\
 &= \frac{\rho g H}{2} \iint \frac{1}{ikG} \frac{\partial \phi_2(r, \theta, \beta)}{\partial r} \Big|_{r=a} \cos \theta dS \\
 &= \frac{2\pi \rho g H a^2}{ikG \cosh(kh)} \sum_{n=1}^N D_n^{0,+} n a^{2n-1} \int_0^1 t P_{2n}^0(t) dt,
 \end{aligned} \quad (33)$$

where we have used the porous boundary condition in Eq. (9), $dS = a^2 \sin \theta d\theta d\beta$, and $\sin \theta = -P_1^1(\cos \theta)$. In Eq. (33), the integral of $\int_0^1 t P_{2n}^0(t) dt$ is calculated by [40, Eq. (7.126-1)]

$$\int_0^1 t P_{2n}^0(t) dt = \frac{\sqrt{\pi}}{4(n+1)! \Gamma(3/2 - n)}, \quad (34)$$

where $\Gamma(x) = \int_0^\infty \tau^{x-1} e^{-\tau} d\tau$ is the Gamma function. The dimensionless horizontal and vertical wave forces are respectively defined as

$$f_x = \frac{|F_x|}{\pi \rho g H a^2}, \quad (35)$$

$$f_z = \frac{|F_z|}{\pi \rho g H a^2}. \quad (36)$$

The free surface elevation is calculated by

case, we only need consider the velocity potential ϕ_1 in the exterior region given by Eq. (23). Then, the porous boundary condition in Eq. (9) becomes

$$\frac{\partial \phi_1}{\partial r} = 0, \quad r = a. \quad (39)$$

The system of linear equations formulated by Eqs. (27)–(30) is reduced to

$$\begin{aligned}
 C_j^{m,+} - \sum_{n=0}^{\infty} C_n^{m,+} \frac{(2j+2m)a^{2n+2j+4m+1} A_{nj}^{m,+}}{2j+2m+1} \\
 = \frac{2j+2m}{2j+2m+1} \frac{\varepsilon_m (-1)^m (ka)^{2j+2m}}{(2j+4m)! a}, \quad m = 0, 1, 2, \dots, \\
 j = 0, 1, 2, \dots,
 \end{aligned} \quad (40)$$

$$\begin{aligned}
 C_j^{m,-} - \sum_{n=0}^{\infty} C_n^{m,-} \frac{(2j+2m-1)a^{2j+2n+4m-1} A_{nj}^{m,-}}{2j+2m} \\
 = \frac{2j+2m-1}{2j+2m} \frac{2i(-1)^m (ka)^{2j+2m-1}}{(2j+4m-2)! a}, \quad m = 1, 2, \dots, \\
 j = 0, 1, 2, \dots
 \end{aligned} \quad (41)$$

Solving the system of linear equations formulated by Eqs. (40) and (41) after truncating the series gives the unknown coefficients in the velocity potential ϕ_1 .

$$\eta(x, y) = \frac{H}{2} \phi_1 = \frac{H}{2} e^{ikx} + \frac{H}{2 \cosh(kh)} \left\{ \sum_{m=0}^M \sum_{n=0}^N C_n^{m,+} a^{2n+2m+2} \phi_{n+m}^{m,+} \Big|_{z=0} \cos(2m\beta) \right. \\
 \left. + \sum_{m=1}^M \sum_{n=0}^N C_n^{m,-} a^{2n+2m+1} \phi_{n+m}^{m,-} \Big|_{z=0} \cos((2m-1)\beta) \right\}, \quad (37)$$

where the values of $\phi_{n+m}^{m,+} \Big|_{z=0}$ and $\phi_{n+m}^{m,-} \Big|_{z=0}$ are evaluated using the original expressions in Eqs. (15) and (16). The dimensionless free surface elevation is defined as

$$C_H = \frac{2|\eta|}{H}. \quad (38)$$

4 The special case of an impermeable reef ball

When the porous effect parameter G equals zero, the reef ball will become impermeable. For this special

Now the solution is reduced to that in Chapman [21]. Actually, our calculation results of “scattering coefficient” for this special case are exactly the same as that of Chapman [21]. The horizontal and vertical wave forces acting on the impermeable reef ball are respectively calculated by

$$\begin{aligned}
 F_x &= \frac{\rho g H}{2} \iint \phi_1(a, \theta, \beta) (-\sin \theta \cos \beta) dS \\
 &= \frac{\pi \rho g H a^3}{\cosh(kh)} C_0^{1,-},
 \end{aligned} \quad (42)$$

$$F_z = \frac{\rho g H}{2} \iint \phi_1(a, \theta, \beta)(-\cos \theta) dS = \frac{-\pi \rho g H a^2}{2 \cosh(kh)} \left[1 + \sum_{n=1}^N C_n^{0,+} \left(4 + \frac{1}{n} \right) a \int_0^1 t P_{2n}^0(t) dt + \sum_{n=1}^N C_n^{0,+} a^{2n+2} A_{n0}^{0,+} \right], \quad (43)$$

where we have used Eqs. (40) and (41). The free surface elevation is still calculated by Eq. (37).

5 Results and discussions

5.1 Validations

For the systems of linear equations in Eqs. (27)–(30) or Eqs. (40) and (41), we have two truncation parameters which must be considered carefully. The first one is the maximum value N , which determines the convergence of the wave forces acting on the reef ball. The second one is the maximum value M in the angle β mode, which determines the accuracy of the free surface elevation η , as indicated in Eq. (37). Therefore, we carry out some calculation examples to examine the convergence of the present solution with the increasing truncated numbers. The calculation results of the dimensionless horizontal and vertical wave forces, f_x and f_z , acting on the submerged reef ball at different truncated numbers N are listed in Tables 1, 2 and 3. It can be seen from these tables that the convergence of the results is very rapid, and using a truncated number $N = 7$ can yield results with five-figure accuracy. Higher accurate results can be obtained by using a larger truncated number N . Besides, the convergence of the dimensionless free surface elevation C_H at the points $(x, y, z) = (h, 0, 0)$ and $(-h, 0, 0)$ with different truncated numbers M (the number of angle β modes is $2M + 1$) are examined in Tables 4 and 5. Here the value of N equals seven. It can be seen from Tables 4 and 5 that the convergence of the values of C_H are also very rapid, and the calculation results with five-figure accuracy can be obtained by using the truncated number $M = 4$. Thus, the values of $N = 7$ and $M = 4$ will be used in the subsequent calculations.

For the present wave scattering problem, we have also developed a three-dimensional multi-domain BEM solution for cross-checking of the analytical

solution. The solution procedure of the multi-domain BEM is detailed in “Appendix”. We note from Eq. (10) that the porous effect parameter G may vary with the incident wave number k . Therefore, an alternative normalized porous effect parameter $G_0 = h\varepsilon/[\delta(f - is)] = Gkh$ is used in this study, which has been adopted by Liu and Li [43]. It is noted that G (G_0) is a complex quantity and its real and imaginary parts denote the resistance and inertial effects of the perforated plate (perforated hemispherical shell), respectively [32]. However, the inertial effect may be not significant for the wave scattering problem in practice [34, 44]. Therefore, the present calculations will only consider a real G_0 . Comparisons between the typical calculation results by two different solutions are shown in Fig. 2. In the multi-domain BEM solution, 1728 quadrilateral elements on the surface of the reef ball are used for both cases in Fig. 2. As an example, the meshes of the multi-domain BEM solution for the case of Fig. 2a is plotted in Fig. 3. The lines and dots in Fig. 2 denote the results of the analytical solution and the multi-domain BEM solution, respectively. It can be seen from this figure that the results of two different solutions are in good agreement, which means that both solutions should be reliable.

5.2 Wave forces

Figures 4 and 5 give the variations of the dimensionless horizontal and vertical wave forces, f_x and f_z , with wave number kh for a submerged impermeable reef ball at different a/h , respectively. Here, four values of a/h from 0.4 to 0.7 are considered. It can be seen from Fig. 4 that with the increasing value of kh , the value of f_x first increases, reaches a maximum value and then decreases. The maximum value of f_x increases with the increasing value of reef ball radius a/h . It is noted from Fig. 5 that the value of f_z decreases gradually with the increasing value of kh for all the cases of a/h , and tends to be zero eventually. It is also noted from Fig. 5 that

Table 1 Convergence examination of the dimensionless wave forces on an impermeable reef ball at $a/h = 0.5$

Truncated number N	$kh = 0.5$		$kh = 1.0$		$kh = 1.5$		$kh = 2.0$	
	f_x	f_z	f_x	f_z	f_x	f_z	f_x	f_z
0	0.11376	0.44341	0.16549	0.32403	0.15944	0.21255	0.12994	0.13290
1	0.11377	0.44972	0.16557	0.34239	0.15962	0.23806	0.13009	0.15894
2	0.11377	0.44972	0.16557	0.34232	0.15962	0.23785	0.13009	0.15853
3	0.11377	0.44972	0.16557	0.34232	0.15962	0.23785	0.13009	0.15854
4	0.11377	0.44972	0.16557	0.34232	0.15962	0.23785	0.13009	0.15854
5	0.11377	0.44972	0.16557	0.34232	0.15962	0.23785	0.13009	0.15854
6	0.11377	0.44972	0.16557	0.34232	0.15962	0.23785	0.13009	0.15854
7	0.11377	0.44972	0.16557	0.34232	0.15962	0.23785	0.13009	0.15854
8	0.11377	0.44972	0.16557	0.34232	0.15962	0.23785	0.13009	0.15854

Table 2 Convergence examination of the dimensionless wave forces on an impermeable reef ball at $a/h = 0.8$

Truncated number N	$kh = 0.5$		$kh = 1.0$		$kh = 1.5$		$kh = 2.0$	
	f_x	f_z	f_x	f_z	f_x	f_z	f_x	f_z
0	0.19793	0.44341	0.28101	0.32403	0.25114	0.21255	0.19197	0.13290
1	0.19869	0.46565	0.28554	0.39089	0.25874	0.28647	0.19510	0.17792
2	0.19872	0.46560	0.28568	0.39096	0.25918	0.28653	0.19584	0.17433
3	0.19872	0.46562	0.28568	0.39107	0.25919	0.28682	0.19587	0.17477
4	0.19872	0.46562	0.28568	0.39106	0.25919	0.28681	0.19587	0.17474
5	0.19872	0.46562	0.28568	0.39106	0.25919	0.28681	0.19587	0.17475
6	0.19872	0.46562	0.28568	0.39106	0.25919	0.28681	0.19587	0.17475
7	0.19872	0.46562	0.28568	0.39106	0.25919	0.28681	0.19587	0.17475
8	0.19872	0.46562	0.28568	0.39106	0.25919	0.28681	0.19587	0.17475

Table 3 Convergence examination of the dimensionless wave forces on a perforated reef ball at $a/h = 0.9$ and $G = 0.5$

Truncated number N	$kh = 0.5$		$kh = 1.0$		$kh = 1.5$		$kh = 2.0$	
	f_x	f_z	f_x	f_z	f_x	f_z	f_x	f_z
1	0.21402	0.022779	0.24041	0.066186	0.17594	0.073065	0.11059	0.057220
2	0.21419	0.022623	0.24073	0.065636	0.17612	0.069752	0.10965	0.050192
3	0.21421	0.022728	0.24076	0.066087	0.17611	0.070253	0.10951	0.050503
4	0.21421	0.022722	0.24077	0.066057	0.17611	0.070149	0.10949	0.050322
5	0.21421	0.022725	0.24077	0.066071	0.17611	0.070167	0.10948	0.050338
6	0.21421	0.022725	0.24077	0.066069	0.17611	0.070161	0.10948	0.050333
7	0.21421	0.022725	0.24077	0.066069	0.17611	0.070162	0.10948	0.050331
8	0.21421	0.022725	0.24077	0.066069	0.17611	0.070162	0.10948	0.050331

the value of f_z tends to 0.5 when kh approaches to zero, which is totally different from the horizontal wave force. However, the effect of a/h on f_z is similar to that on f_x .

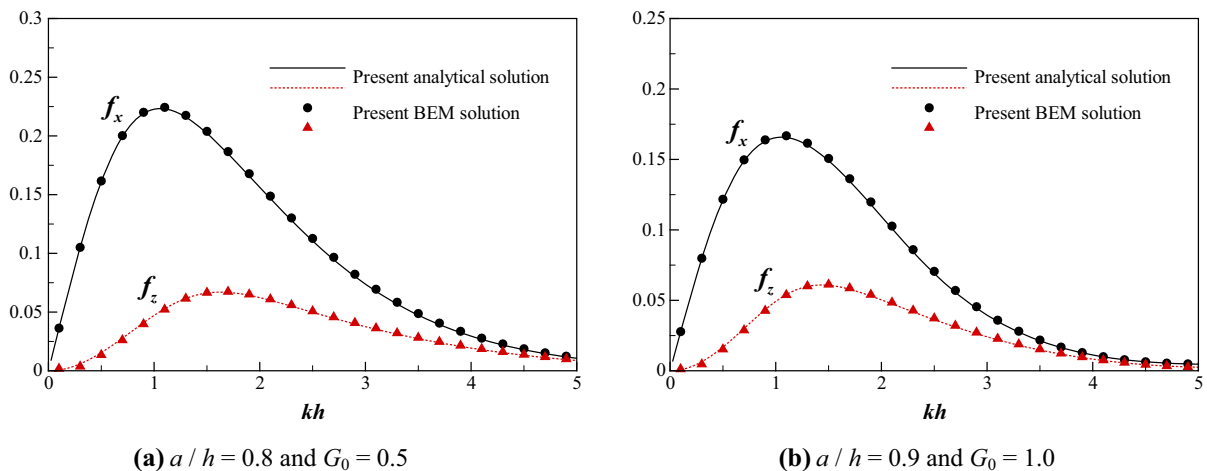
The variations of the dimensionless horizontal and vertical wave forces, f_x and f_z , with wave number kh for perforated reef ball with different a/h are shown in Figs. 6 and 7, respectively. A value of $G_0 = 0.5$ is used

Table 4 Convergence examination of the dimensionless free surface elevation C_H near an impermeable reef ball at $a/h = 0.8$

Truncated number M	$kh = 0.5$		$kh = 1.0$		$kh = 1.5$		$kh = 2.0$	
	$(h, 0, 0)$	$(-h, 0, 0)$	$(h, 0, 0)$	$(-h, 0, 0)$	$(h, 0, 0)$	$(-h, 0, 0)$	$(h, 0, 0)$	$(-h, 0, 0)$
1	1.04713	1.09359	1.14684	1.20834	1.25273	1.03291	1.29906	0.95794
2	1.04705	1.09352	1.14585	1.20750	1.24946	1.02967	1.29300	0.95166
3	1.04705	1.09352	1.14585	1.20751	1.24946	1.02968	1.29302	0.95168
4	1.04705	1.09352	1.14585	1.20751	1.24946	1.02968	1.29302	0.95168

Table 5 Convergence examination of the dimensionless free surface elevation C_H near a perforated reef ball at $a/h = 0.9$ and $G = 0.5$

Truncated number M	$kh = 0.5$		$kh = 1.0$		$kh = 1.5$		$kh = 2.0$	
	$(h, 0, 0)$	$(-h, 0, 0)$	$(h, 0, 0)$	$(-h, 0, 0)$	$(h, 0, 0)$	$(-h, 0, 0)$	$(h, 0, 0)$	$(-h, 0, 0)$
1	1.00441	1.16680	1.00488	1.11677	0.97169	0.95432	0.93398	0.98325
2	1.00425	1.16660	1.00273	1.11469	0.96399	0.94693	0.91852	0.97391
3	1.00425	1.16660	1.00274	1.11469	0.96402	0.94696	0.91859	0.97399
4	1.00425	1.16660	1.00274	1.11469	0.96402	0.94696	0.91859	0.97399


Fig. 2 Comparisons between the analytical solution and the multi-domain BEM solution

in the two figures. It can be seen from Fig. 6 that the variation of f_x with the increasing value of kh is similar to that of impermeable reef balls. However, we note from Fig. 7 that different from impermeable reef balls, the value of f_z first increases, then reaches a peak value, and finally decreases gradually with the increasing value of kh . Moreover, perforating the reef ball can reduce the wave forces significantly by comparing Figs. 4 and 5 with Figs. 6 and 7, which

may be benefit to enhancing the safety and stability against wave action.

Figures 8 and 9 show the variations of the dimensionless horizontal and vertical wave forces, f_x and f_z , with wave number kh at $a/h = 0.5$ and different G_0 , respectively. Here five different values of the porous effect parameter ($G_0 = 0.0, 0.5, 1.0, 1.5$ and 2.0) are considered. We note from Fig. 8 that with different values of G_0 , the variations of the horizontal wave

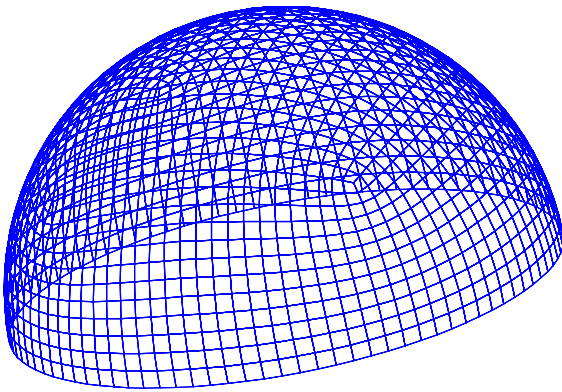


Fig. 3 Meshes of the multi-domain BEM solution for the case of Fig. 2a

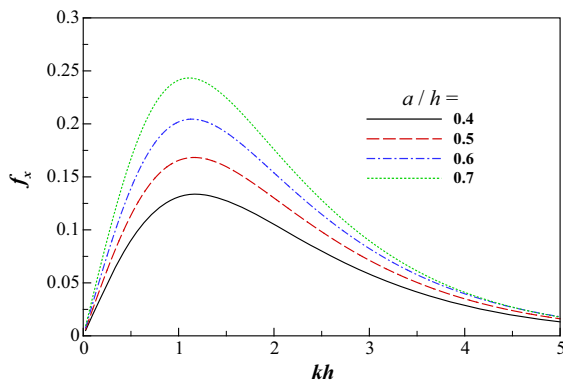


Fig. 4 Variation of the dimensionless horizontal wave force f_x with kh for the impermeable reef ball at different a/h

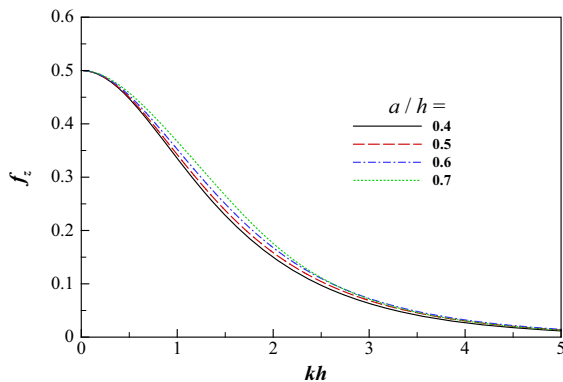


Fig. 5 Variation of the dimensionless vertical wave force f_z with kh for the impermeable reef ball at different a/h

force f_x against wave number kh are similar, and all the curves of f_x reach maximum values at $kh \approx 1.2$. We note from Fig. 9 that the value of f_z for the

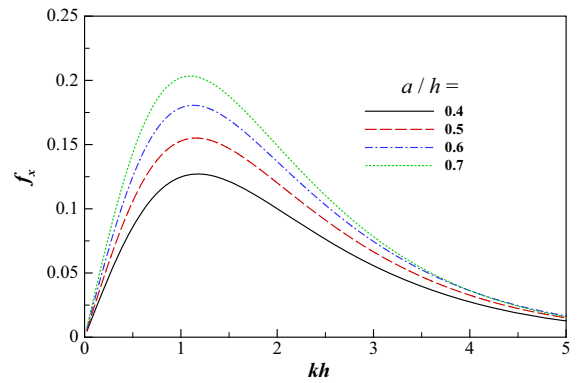


Fig. 6 Variation of the dimensionless horizontal wave force f_x with kh for the perforated reef ball at $G_0 = 0.5$ and different a/h

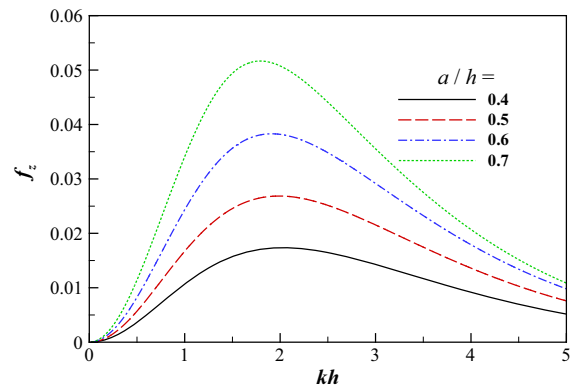


Fig. 7 Variation of the dimensionless vertical wave force f_z with kh for the perforated reef ball at $G_0 = 0.5$ and different a/h

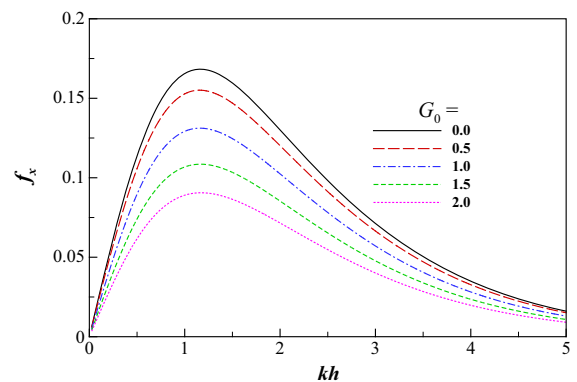


Fig. 8 Variation of the dimensionless horizontal wave force f_x with kh at $a/h = 0.5$ and different G_0

impermeable reef ball ($G_0 = 0.0$) remarkably differs from that for the perforated reef ball ($G_0 \neq 0.0$) when

the value of kh tends to zero, i.e. the incident wavelength becomes very large. This is due to the fact that when the wavelength is very large, the velocity potentials on the external and internal surfaces of the perforated reef ball are almost consistent. That is to say, there is no dynamic pressure difference between both surfaces, and thus the value of f_z tends to zero. But for the impermeable reef ball, the velocity potential on the internal surface always equals zero, and thus the value of f_z is non-zero when the value of kh tends to zero. It is observed from Fig. 9 that the value of f_z on an impermeable reef ball is significantly larger than that of the corresponding perforated one. From Fig. 9, f_z reaches a maximum value at $kh \approx 2.0$ for all the perforated cases.

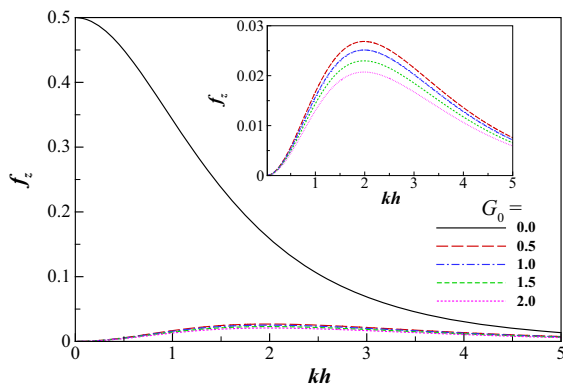


Fig. 9 Variation of the dimensionless vertical wave force f_z with kh at $a/h = 0.5$ and different G_0

Moreover, the maximum values of f_x and f_z on the perforated reef ball decrease significantly with the increasing value of G_0 from 0.5 to 2.0. In Figs. 8 and 9, the effects of G_0 on wave forces are examined in the whole wave frequency range and only several values of G_0 are considered. In order to further investigate the effects of G_0 on the wave forces, we plot the dimensionless horizontal and vertical wave forces f_x and f_z as functions of G_0 at $a/h = 0.5$ and $kh = 0.5$ and 1.0 in Fig. 10. It is evident from this figure that both values of f_x and f_z decrease with the increasing value of G_0 .

Figure 11 gives the variation of the phase difference C_θ between the horizontal and vertical wave forces on the impermeable reef ball at different a/h . It is worth noting that F_x and F_z are in-phase when

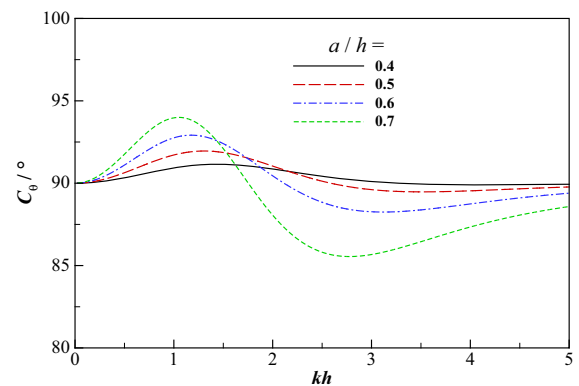


Fig. 11 Variation of the phase difference C_θ between F_x and F_z with kh for the impermeable reef ball at different a/h

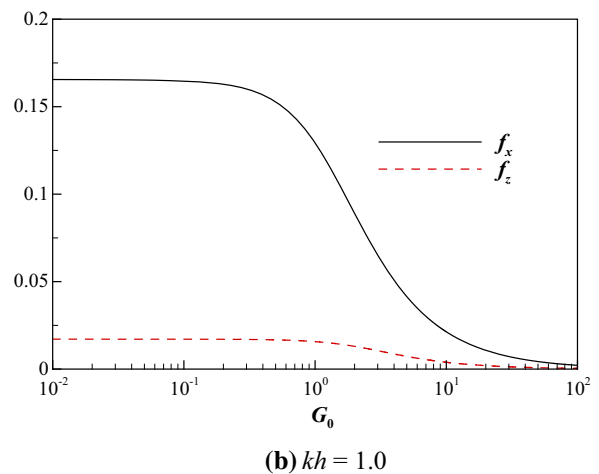
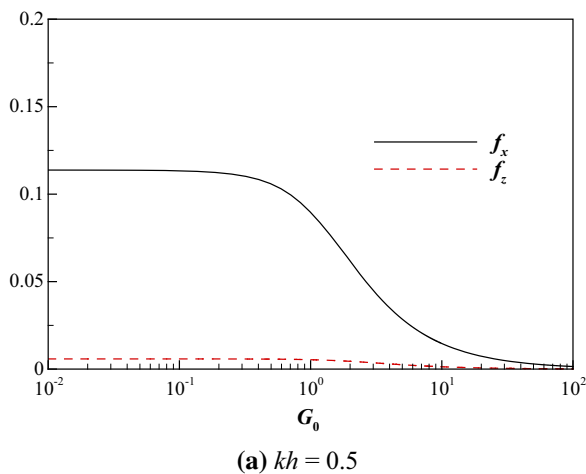


Fig. 10 Variations of the dimensionless horizontal and vertical wave forces, f_x and f_z , with G_0 at $a/h = 0.5$

$C_\theta = 0^\circ$, and are out-phase when $C_\theta = 180^\circ$. Out-phase indicates that when the horizontal wave force attains the maximum, the vertical wave force also attains its maximum and points to downwards. Thus, a larger value of C_θ means that the reef ball has a better stability against sliding. From Fig. 11, there is an obvious phase difference between the horizontal and vertical wave forces, which is helpful for enhancing the stability against the sliding of the reef ball. We note from Fig. 11 that in whole wave frequency, the variation range of the value of C_θ is wider for a larger value of a/h . Figure 12 shows the variation of the phase difference C_θ between the horizontal and vertical wave forces on the perforated reef ball at $a/h = 0.5$ and different G_0 . In comparison with an impermeable reef ball, the value of C_θ for the corresponding perforated one is larger. This may be beneficial to enhancing the stability against sliding for the perforated reef ball.

5.3 Free surface elevation

In preceding calculation examples, the horizontal and vertical wave forces acting on the submerged reef ball are examined. Here, the variations of the dimensionless free surface elevation C_H against wave number kh at a large relative radius $a/h = 0.7$ and different G_0 are shown in Fig. 13. The values of C_H at four different points of $(x, y, z) = (-h, 0, 0)$, $(0, h, 0)$, $(h, 0, 0)$ and $(0, 0, 0)$ are calculated. From Fig. 13, the transformation of waves above the reef ball is significant. It can be observed from Fig. 13a, b that peak and trough occur in the curves of C_H with the increasing value of

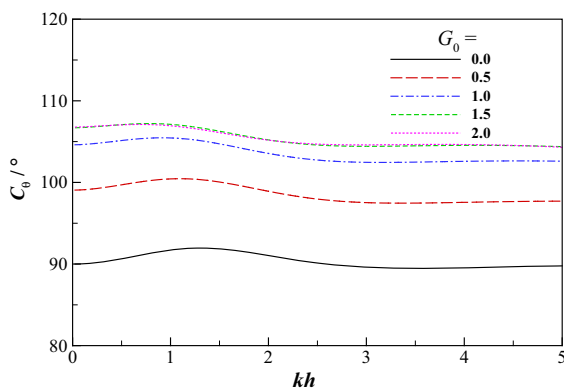


Fig. 12 Variation of the phase difference C_θ between F_x and F_z with kh for the perforated reef ball at $a/h = 0.5$ and different G_0

kh , and when the value of G_0 increases, the peak value of C_H decreases significantly and the value of C_H approaches to unity gradually. It is noted from Fig. 13c, d that in the case of short incident waves, the values of C_H at $(x, y, z) = (h, 0, 0)$ and $(0, 0, 0)$ decrease when the value of G_0 increases from 0 to 4.0. We also note that when the value of kh approaches to zero, the values of C_H tends to unity for all the cases.

Finally, Figs. 14, 15 and 16 give the contours of the dimensionless free surface elevation C_H near the submerged reef ball at $a/h = 0.7$ and $kh = 0.5, 1.0$ and 3.0 . An impermeable reef ball and two corresponding perforated ones with $G_0 = 0.5$ and 1.0 are examined. The red dashed circles in Figs. 14, 15 and 16 denote the position of the submerged reef ball. It can be seen from these figures that for the cases of $kh = 0.5$ and 1.0 , the effect of the reef ball on the free surface elevation is notable for both seaside and leeside regions; however, almost no effect for seaside region is observed in the case of $kh = 3.0$ (short wave). We also note that the transformation of free surface elevation for a perforated reef ball is gentler than that of the corresponding impermeable one, and the effect of the reef ball on the transformation of free surface elevation becomes smaller when the value of G_0 increases. By comparing Figs. 14, 15 and 16, the transformation of free surface elevation in leeside region is more complicated and intense for shorter waves (larger value of kh) compared with that for longer waves.

6 Limitation of the present solution

In practical situations, high-order harmonic waves may be produced when water waves propagate over a submerged breakwater [45]. At that case, the nonlinearity of the free surface cannot be considered in this study as the present solution is developed on the basis of the linear potential theory. In addition, the present solution also cannot consider wave breaking over perforated reef ball.

In real fluid, the vortexes and the wave energy dissipation are generated by the holes on the reef ball. However, the local flow details near the holes cannot be modeled in present linear potential solution. In this solution, the thin spherical shell is assumed to be a homogenous porous medium (the shell has numerous fine holes). Then, a pressure drop condition [32]

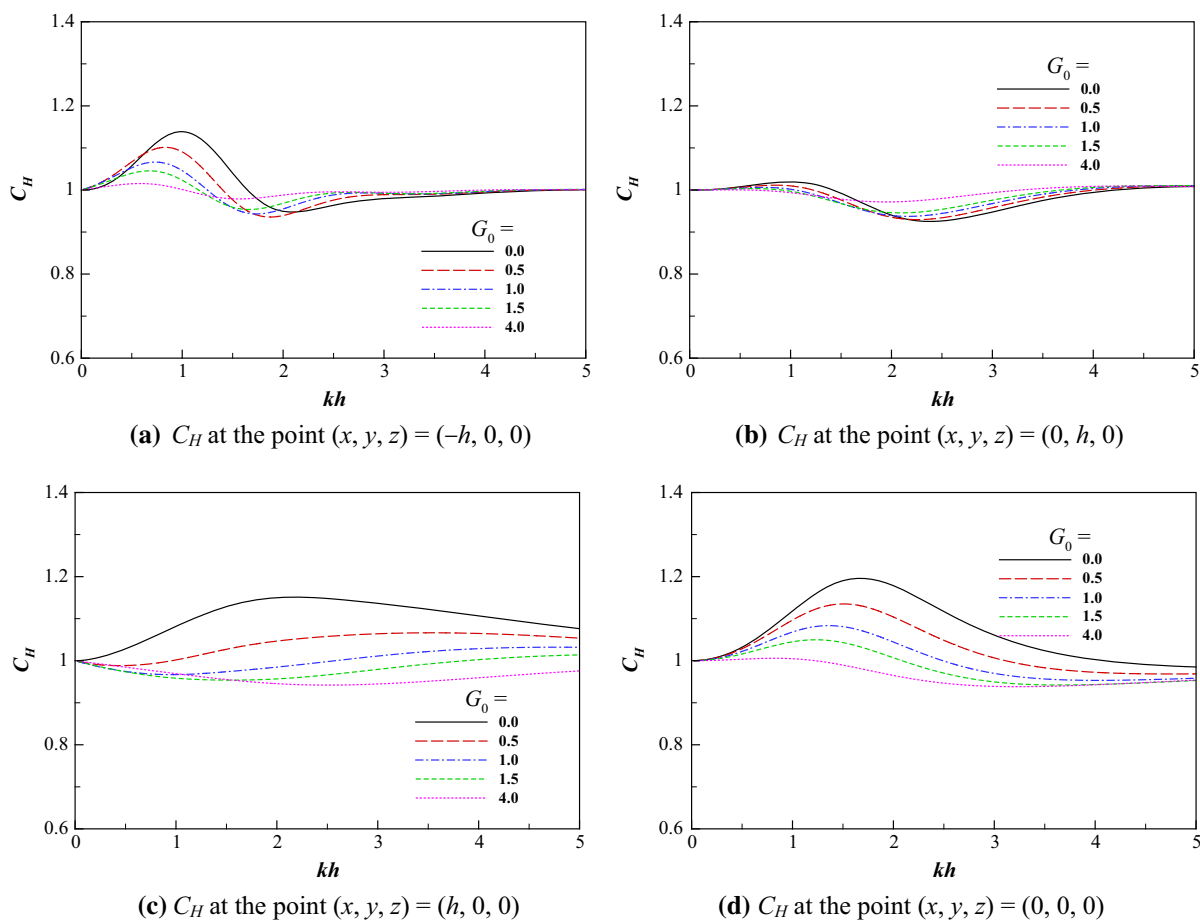


Fig. 13 Variation of the dimensionless free surface elevation C_H with kh at $alh = 0.7$

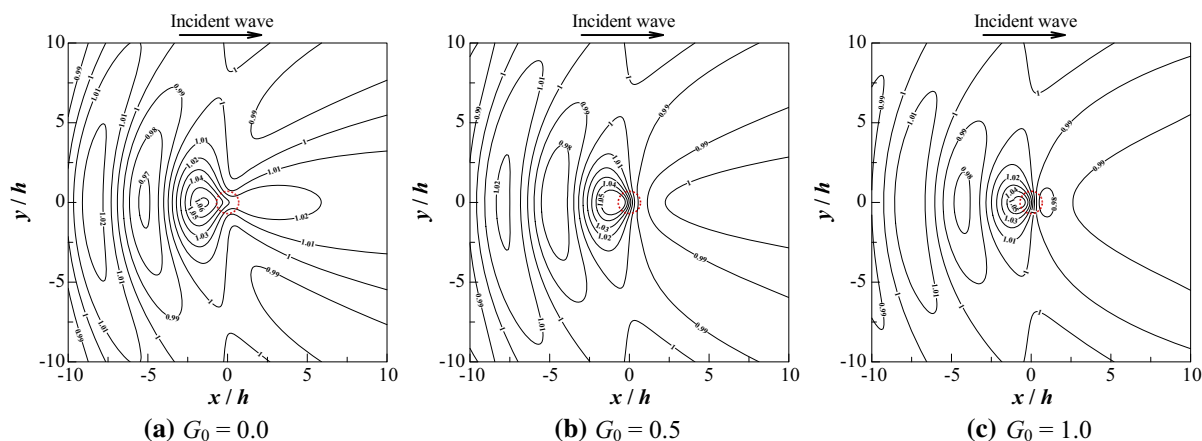
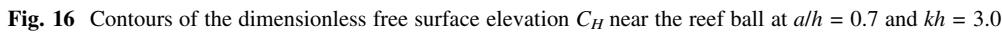
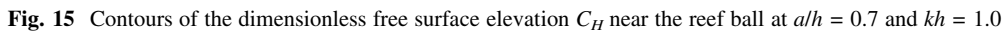


Fig. 14 Contours of the dimensionless free surface elevation C_H near the reef ball at $alh = 0.7$ and $kh = 0.5$

between the fluid velocity and the pressure jump across the perforated shell is used to model the incident wave energy dissipation and the phase

difference caused by the perforated shell. In practical application, the value of the porous effect parameter G in the pressure drop condition (9) must be



of perforated reef balls used as a submerged breakwater through the conversion technique among different spherical coordinate systems [17, 46, Eq. (291)].

Based on linear potential flow theory, an analytical solution for water wave scattering by a submerged perforated reef ball has been developed using the multipole expansion method in this study. The special case of an impermeable reef ball has also been examined. The calculation methods for the horizontal and vertical wave forces acting on the reef ball and the free surface elevations near the reef ball have been

given. A numerical solution for the present wave scattering problem has also been developed based on the three-dimensional multi-domain BEM. The convergence of the present analytical solution has been proved to be very rapid, and the calculation results with five-figure accuracy have been obtained using the truncated numbers of $N = 7$ and $M = 4$. The calculation results of the present analytical solution and the BEM solution have been compared for cross-checking, and the agreement between them was very good.

Typical calculation examples have been presented to examine the basic features of the wave forces on the reef ball and the free surface elevations. It has been found that perforating the hemispherical shell can significantly reduce the wave forces acting on the reef ball, and increase the phase difference between the horizontal and vertical wave forces, which may be beneficial to enhancing the stability of the reef ball against sliding. The variation of horizontal wave force with wave number for the perforated reef ball is similar to that for the impermeable one; however, the basic features of the vertical wave forces on the perforated and impermeable reef balls are totally different. Moreover, perforating the shell can slow down the transformation of the free surface elevation near the reef ball. The transformation of free surface elevation in leeside region is more complicated and intense for shorter waves compared with that for longer waves.

Acknowledgements This study was supported by the Natural Science Foundation of China under Grant Numbers 51725903 and 51490675.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Appendix: Three-dimensional multi-domain BEM solution

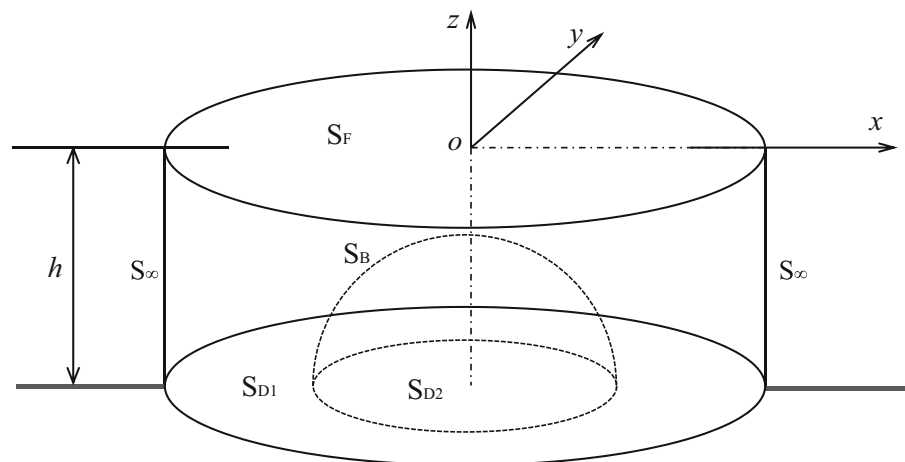
The whole fluid domain is still divided into two regions, which are the same as those in Sect. 2. As shown in Fig. 17, the closed boundary of region 1 is defined as: $S_1 = S_F + S_{D1} + S_B + S_\infty$, where S_F , S_D , S_B and S_∞ are respectively the free surface boundary, the water bottom boundary, the surface boundary of the reef ball and the far field boundary. The closed boundary of region 2 is defined as: $S_2 = S_{D2} + S_B$.

Applying the second Green theorem, the solution of Laplace equation can be converted into the solution of the following boundary integral equation [47]:

$$\begin{aligned} \alpha_j(\mathbf{x}_0)\phi_j(\mathbf{x}_0) &= \iint_{S_j} \left[\phi_j(\mathbf{x}) \frac{\partial \tilde{G}_j(\mathbf{x}; \mathbf{x}_0)}{\partial n_j} - \tilde{G}_j(\mathbf{x}; \mathbf{x}_0) \frac{\partial \phi_j(\mathbf{x})}{\partial n_j} \right] dS_j(\mathbf{x}), \\ j &= 1, 2, \end{aligned} \quad (44)$$

where $\mathbf{x} = (x, y, z)$ and $\mathbf{x}_0 = (x_0, y_0, z_0)$ are the field and source points, respectively; $\alpha_j(\mathbf{x}_0)$ is the interior angle; $\mathbf{n}_j$ denotes the unit normal vector on the boundary S_j and points away from fluid region j ; and $\tilde{G}_j(\mathbf{x}; \mathbf{x}_0)$ is the Green function, which satisfies

Fig. 17 The sketch of three-dimensional multi-domain BEM solution



$$\nabla^2 \tilde{G}_j = \delta(\mathbf{x} - \mathbf{x}_0). \quad (45)$$

For the fluid region 1, the Green function is chosen as [48]:

$$\begin{aligned} \tilde{G}_1(\mathbf{x}; \mathbf{x}_0) = & -\frac{1}{4\pi} \left(\frac{1}{r_0} + \frac{1}{r_1} \right) + \frac{1}{2\pi} \\ & \times \int_0^\infty \frac{(K + \mu)e^{-\mu h} \cosh(\mu(z+h)) \cosh(\mu(z_0+h))}{K \cosh(\mu h) - \mu \sinh(\mu h)} J_0(\mu R_0) d\mu, \end{aligned} \quad (46)$$

where $r_0 = [R_0^2 + (z - z_0)^2]^{1/2}$, $r_1 = [R_0^2 + (z + z_0 + 2h)^2]^{1/2}$ and $R_0^2 = (x - x_0)^2 + (y - y_0)^2$. The path of the integration on the right hand of Eq. (46) passes below the point at $\mu = k$. We split ϕ_1 as $\phi_1 = \phi_0 + \phi_d$ with ϕ_0 being the incident velocity potential given in Eq. (8), and ϕ_d being the diffraction velocity potential to be determined. We note that both the Green function $\tilde{G}_1$ and ϕ_d satisfy Eqs. (4)–(6). Thus, Eq. (44) can be reduced to

$$\begin{aligned} \alpha_1(\mathbf{x}_0)\phi_1(\mathbf{x}_0) = & \iint_{S_B} \left[\phi_1(\mathbf{x}) \frac{\partial \tilde{G}_1(\mathbf{x}; \mathbf{x}_0)}{\partial \mathbf{n}_1} \right. \\ & \left. - \tilde{G}_1(\mathbf{x}; \mathbf{x}_0) \frac{\partial \phi_1(\mathbf{x})}{\partial \mathbf{n}_1} \right] dS_B(\mathbf{x}) + \phi_0(\mathbf{x}_0). \end{aligned} \quad (47)$$

For the fluid region 2, the Green function only satisfying the water bottom condition is given by

$$\tilde{G}_2(\mathbf{x}; \mathbf{x}_0) = -\frac{1}{4\pi} \left(\frac{1}{r_0} + \frac{1}{r_1} \right). \quad (48)$$

Then, Eq. (44) is reduced to

$$\begin{aligned} \alpha_2(\mathbf{x}_0)\phi_2(\mathbf{x}_0) = & \iint_{S_B} \left[\phi_2(\mathbf{x}) \frac{\partial \tilde{G}_2(\mathbf{x}; \mathbf{x}_0)}{\partial \mathbf{n}_2} \right. \\ & \left. - \tilde{G}_2(\mathbf{x}; \mathbf{x}_0) \frac{\partial \phi_2(\mathbf{x})}{\partial \mathbf{n}_2} \right] dS_B(\mathbf{x}). \end{aligned} \quad (49)$$

The surface of the reef ball S_B is divided into I elements. Both velocity potential and its normal derivative on each element are assumed to be constants. Then, all the interior angles equal to 0.5. Now, Eqs. (47) and (49) can be, respectively, discretized into the following linear equations:

$$[\mathbf{A}](\phi_1^1, \phi_1^2, \dots, \phi_1^I)^T + [\mathbf{B}](\bar{\phi}_1^1, \bar{\phi}_1^2, \dots, \bar{\phi}_1^I)^T = \{\mathbf{f}\}, \quad (50)$$

$$[\mathbf{C}](\phi_2^1, \phi_2^2, \dots, \phi_2^I)^T + [\mathbf{D}](\bar{\phi}_2^1, \bar{\phi}_2^2, \dots, \bar{\phi}_2^I)^T = \mathbf{0}, \quad (51)$$

where ϕ_j^m and $\bar{\phi}_j^m$ are respectively the velocity potential and its normal derivative on the center of m th element of the body surface in region j . The matrix coefficients in Eqs. (50) and (51) are given by

$$\begin{aligned} A_{mn} = & \iint_{S_B^n} [\partial \tilde{G}_1(\mathbf{x}^n; \mathbf{x}_0^m) / \partial \mathbf{n}_1] dS_B^n(\mathbf{x}) - 0.5\delta_{mn}, \\ & m, n = 1, 2, \dots, I, \end{aligned} \quad (52)$$

$$\begin{aligned} B_{mn} = & -\iint_{S_B^n} \tilde{G}_1(\mathbf{x}^n; \mathbf{x}_0^m) dS_B^n(\mathbf{x}), \\ & m, n = 1, 2, \dots, I, \end{aligned} \quad (53)$$

$$f_m = -\phi_0(\mathbf{x}_0^m), \quad m = 1, 2, \dots, I, \quad (54)$$

$$\begin{aligned} C_{mn} = & \iint_{S_B^n} [\partial \tilde{G}_2(\mathbf{x}^n; \mathbf{x}_0^m) / \partial \mathbf{n}_2] dS_B^n(\mathbf{x}) - 0.5\delta_{mn}, \\ & m, n = 1, 2, \dots, I, \end{aligned} \quad (55)$$

$$\begin{aligned} D_{mn} = & -\iint_{S_B^n} \tilde{G}_2(\mathbf{x}^n; \mathbf{x}_0^m) dS_B^n(\mathbf{x}), \\ & m, n = 1, 2, \dots, I, \end{aligned} \quad (56)$$

where $\delta_{mn} = 0$ for $m \neq n$ and $\delta_{mn} = 1$ for $m = n$. When the value of R_0/h is greater than 0.1, the Green function in Eq. (45) is calculated using a series method according to that in John [48]. Otherwise, the Green function in Eq. (45) is calculated using the methods described in Chapman [21, Chapter 4], Linton [41, Eqs. (3.7) and (4.11)] and Liu and Li [42, Appendix].

Applying the porous boundary condition on the hemispherical shell

$$-\frac{\partial \phi_1}{\partial \mathbf{n}_1} = \frac{\partial \phi_2}{\partial \mathbf{n}_2} = ikG(\phi_2 - \phi_1), \quad \text{on } S_B, \quad (57)$$

the system of linear equations in Eqs. (49) and (50) can be simplified as

$$\begin{bmatrix} \mathbf{A} + ik\mathbf{GB} & -ik\mathbf{GB} \\ -ik\mathbf{GD} & \mathbf{C} + ik\mathbf{GD} \end{bmatrix} \begin{Bmatrix} \phi_1 \\ \phi_2 \end{Bmatrix} = \begin{Bmatrix} \mathbf{f} \\ \mathbf{0} \end{Bmatrix}. \quad (58)$$

The velocity potentials on all element of the body surface S_B are determined after solving Eq. (58), and their normal derivatives are determined by Eq. (57).

The horizontal and vertical wave forces are respectively calculated by

$$\begin{aligned} F_x &= \frac{\rho g H}{2} \iint_{S_B} (\phi_2 - \phi_1) n_x dS \\ &= \frac{\rho g H}{2} \sum_{m=1}^I (\phi_2^m - \phi_1^m) n_x^m S_B^m, \end{aligned} \quad (59)$$

$$\begin{aligned} F_z &= \frac{\rho g H}{2} \iint_{S_B} (\phi_2 - \phi_1) n_z dS \\ &= \frac{\rho g H}{2} \sum_{m=1}^I (\phi_2^m - \phi_1^m) n_z^m S_B^m, \end{aligned} \quad (60)$$

where n_x^m and n_z^m are respectively the components along x - and z -directions of the unit normal vector on the m th element pointing to region 2, and S_B^m is the area of the m th element on S_B .

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